



DESIGN SPECIFICATION DOCUMENT

PROJECT: TIGO PESA AGENCY BANKING

Version 1.4

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REVISION HISTORY

Date	Version	Description	Revised By
4 th April 2022	1.0.0	Initial Draft	Geoffrey Njoroge
2 nd February, 2023	1.1	Added cashcode withdrawal service	Geoffrey Njoroge
4 th March, 2023	1.2	Updated to include code generating on Equity channels	Geoffrey Njoroge
5 th March, 2023	1.3	Removed cash deposit using Equity bank channels and only maintained deposit using Tigo Pesa Agent channel	Geoffrey Njoroge
29 th March, 2023	1.4	Add API documentation and change flows to Asynchronous	Geoffrey Njoroge

Sign Off required

Name	Position	Signature	Date
Shadrack Mwagiru	Head of Enterprise Architecture	OK	11 th April 2022



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1. INTRODUCTION

Scope of this document

The purpose of this document is to provide a comprehensive specification for Tigo pesa Agency banking services. The scope covered in the document include:

- a. Equity bank customer withdrawing at Tigo Pesa Agent using Tigo Menu
- b. Equity bank customer withdrawing at Tigo Pesa Agent using Equity bank channels
- c. Depositing to Equity Bank account using Tigo Pesa Agent or Tigo pesa menu
- d. Depositing to Equity bank account using Equity bank channel

2. DESIGN CONSTRAINTS, ASSUMPTIONS, DEPENDENCIES AND RISKS

Design Constraints and Assumptions

Assumptions

#	Description
1.	An API gateway exists. All API calls to and from the Telco must go through the API gateway
2.	The business rules applicable for this solution are available and to be configured on the Telco business engine
3.	All independent services are available
4.	Availability of TigoPesa APIs

Dependencies

- Integration to TigoPesa that will enable transfer of money to and from Equity system
- Integration to Core Banking System (Finacle)
- High availability and good performance of the APIs.

Risks

1. Network connection failure
2. Network latency leading to increased timeouts and reversals.
3. Customer and TigoPesa agent account details validation failure

Logging

Standard Logging guidelines will be followed while implementing logging for the Equity Bank as follows;

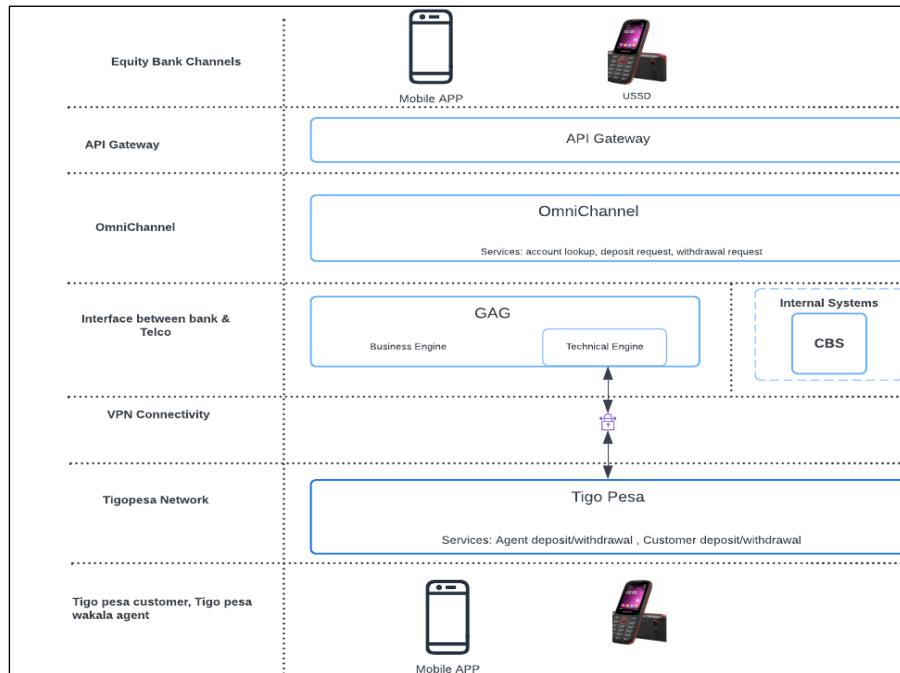
1. The Input, Output and Error messages will be logged.
2. Any sensitive information (e.g. passwords, keys) will not be stored in the logs
3. An audit trail of privileged and user accounts activities will be maintained (including successful and failed login attempts).
4. All system components will have logging in place.
5. Log level will be controlled as a configuration



3. SOLUTION ARCHITECTURE OVERVIEW

High-level architecture

Below is a high-level architectural design of the proposed solution



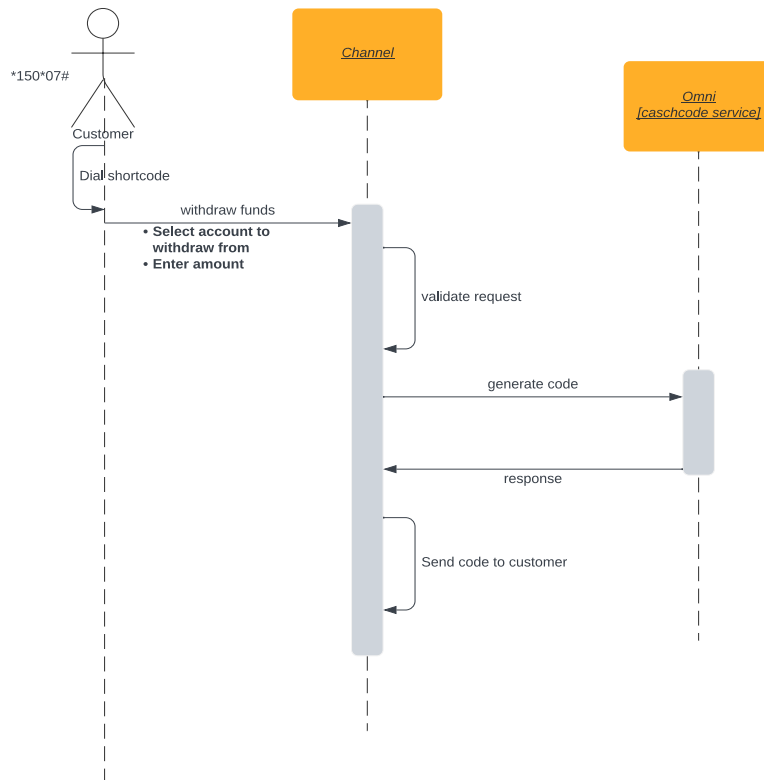
Architectural Description

- Equity bank customers and non-Equity bank customers will be able to deposit/withdraw cash at a Tigo pesa agent either using a bank channel or a Tigopesa channel
- Connection between Tigo pesa and Equity bank systems will be through a secure VPN connection
- Customers will be able to initiate withdrawal either through the bank channels, the request handled by the channel through the API gateway to omni channel. Omnichannel sends the request through GAG to Tigo pesa network
- Similarly, customers will initiate requests through Tigopesa menu / Tigopesa Agent and the request sent to Bank network.
- The Technical service will be the interface facing the Tigo network, whereas the business engine will be the interface facing the internal bank systems
- Omnichannel will manage connection to Equity bank channels

4. INTERFACE REQUIREMENTS

A) GENERATING CASHCODE TO USE FOR WITHDRAWAL

This service will allow Equity bank customers to request a cashcode that will be used to withdraw at the agent. Below is the sequence diagram



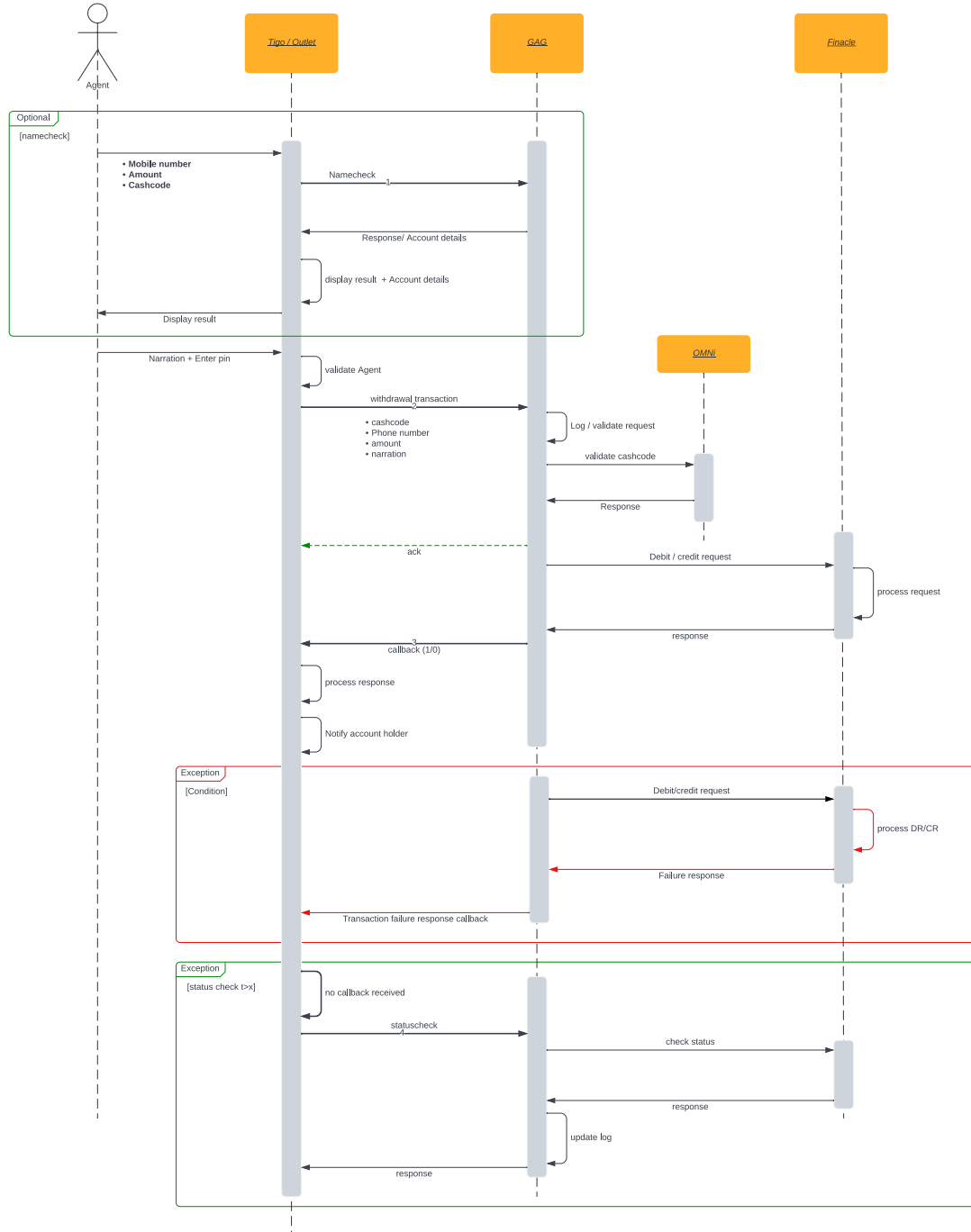
Detailed description

- A customer using Equity bank channels select withdraw funds by code
- Channel receives the request and forwards cashcode service to generate code. A customer will select account to withdraw from and amount to withdraw.
- Business rule will guide cashcode to expire time. This is configurable
- Once the customer receives the code, the customer can initiate withdraw service at any Tigo agent.

B) CASH WITHDRAW MONEY FROM EQUITY BANK ACCOUNT AT TIGO PESA AGENT

This service will allow Equity bank customers to withdraw cash at Tigo Pesa Agent at a Tigo pesa agent after generating the cashcode in step A above.

Below is the sequence diagram:



Detailed description

- After a customer has generated the cashcode, walks in to a Tigopesa agent and asks to withdraw funds from their account.
- Tigopesa agent asks customer to present account details and cashcode to the agent who captures request for withdrawal on Tigo menu



- Mobile number
 - Amount
 - cashcode
 - Narration - optional
- After agents clicks submit, Tigo validates the request and submits to the Bank through GAG for
 - Cashcode validation – If expired / available
 - Account details to withdraw
- Response to Tigo from Bank is the status of the cashcode and account details
- If the response is an ok, the Agent submits and enters pin for confirmation
- The request is sent to the bank for processing.
- After successful response the Agent disburses funds to the customer

Exception (Insufficient funds / Transaction failure or network failure)

- At this point, the transaction will not be processed and the customer will be notified by the Agent of the transaction failure.

Exception – Failure on Tigo side to complete transaction processing:

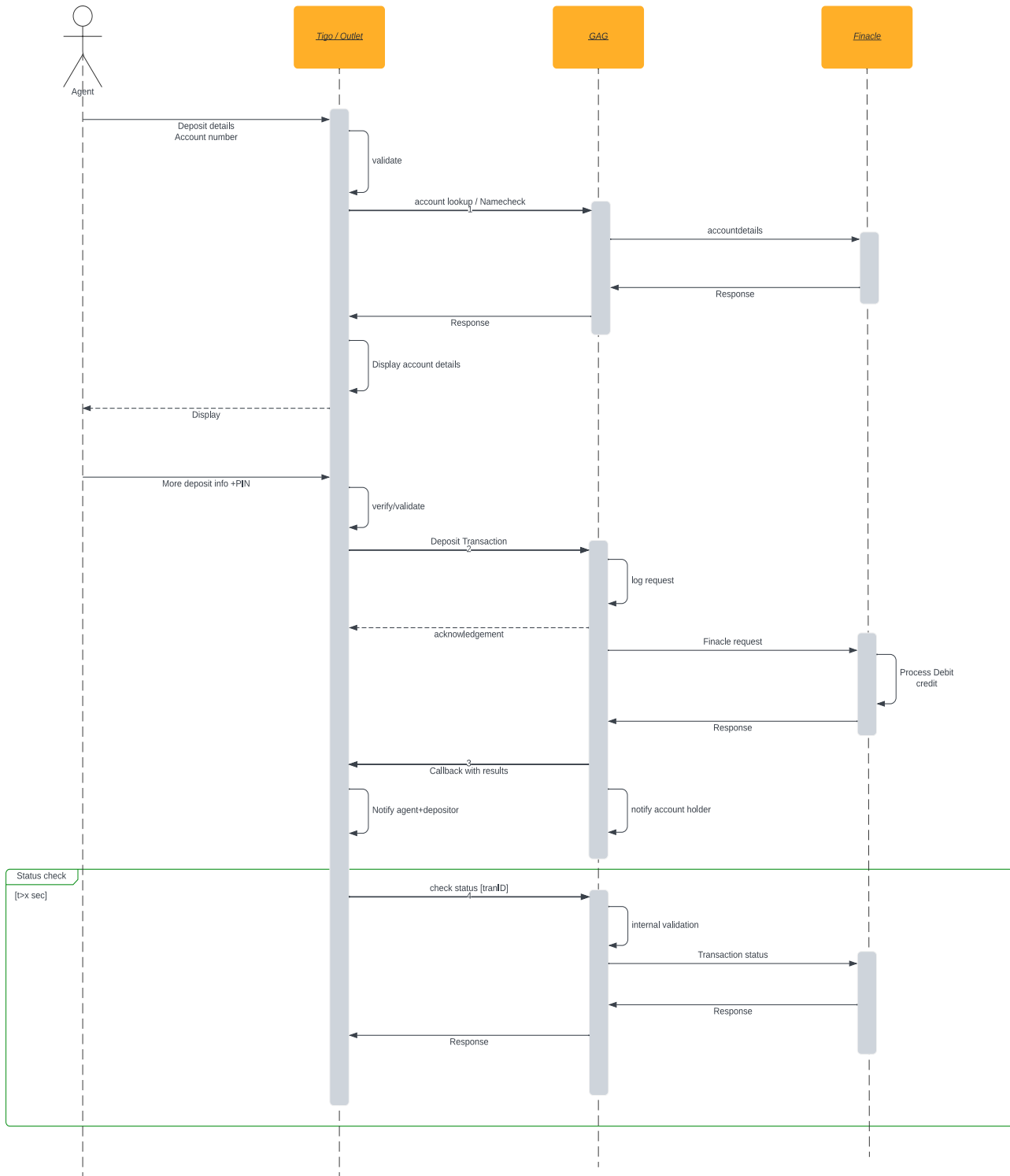
Equity bank will provide an endpoint so that Tigo pesa will call for a rollback if the transaction fails. The bank will process the reversal and also notify the customer of the reversal.



C) DEPOSIT MONEY TO EQUITY BANK ACCOUNT THROUGH TIGOPESA AGENT

This service allows Equity bank customers and non-Equity bank customers deposit cash to an Equity bank account at the TigoPesa Agent.

Below is the sequence diagram for the integration points





Below is the detailed description to the sequence diagram

- Customer walks to a Tigopesa agent and requests to make a deposit to an Equity bank account
- Agent requests customer for account details and customer identity details
- Request is sent to Tigopesa where Tigopesa sends an namecheck (account lookup) request and displays the information to the agent to confirm or cancel
- After successful confirming the request, Tigopesa sends a deposit request to the bank through the technical engine. Technical engine sends a deposit request (C2B) to business engine to credit customer account and debit telco gl. After successful deposit, account holder is notified of the transaction



INTERFACE DEFINITION (GENERIC TO TELCO)

WITHDRAWAL TRANSACTION APIS

This section covers the generic APIs that GAG will expose to the Telco for consumption:

a. Authentication API

HTTP Version	HTTP/1.1
API TYPE	REST/JSON
HTTP Method	POST
End-point	{{baseURL}}/v1/api/oauth/token
Mode	Synchronous

Headers

Key	Required	Value	Description
Content-Type	yes	application/x-www-form-urlencoded	Content Type
Authorization	yes	base64-encoded(username.password)	Basic Auth

Sample request

```
POST /v1/api/oauth/token HTTP/1.1
Host: {{baseURL}}
Authorization: Basic bWdlcnVzaDptZ3VydXNo
Content-Type: application/x-www-form-urlencoded
username={{username}}&password={{password}}&grant_type=password
```

Sample response

```
HTTP code : 200
{
  "access_token":
  "eyJ0eXAiOiJKV1QiLCJhbGciOiJIUzI1NiJ9.eyJpc3MiOiJpbmtpbmUgSldUIEJlaWxkZXIiLCJpYXQiOiJlODcsImV4cCI6MTcxMTczMjU4NywiYXVkIjoibWV4YW1wbGUuY29tIiwic3ViIjoibG9rZW4gRXhjbXBsZSI6InVzZXJfbmFtZSI6ImlRJR09fVGFoiLCJhdXRob3JpdGllcyI6WyJBRElJTlIsIkFETU10sImNsYWVudF9pZCI6InRpdz29fdHoiLCJzY29wZSI6WyJyZWZkIiwid3JpdGUuXX0.gYMtp7mOF0_cBvZ_K12OagbyCHrWkWaQKwTtt1bjxWE",
  "token_type": "bearer",
  "expires_in": 19999,
  "scope": "read write",
  "jti": "mzk5loEMhfg4jP4WGx-Ijde7JyI"
}
```



b. Namecheck

This service will be used for account lookup (Namecheck). The response will contain account details of the mapped phone number.

HTTP Version	HTTP/1.1
API TYPE	REST/JSON
HTTP Method	POST
End-point	{{baseUrl}}/api/v1/withdraw/namecheck
Mode	Synchronous

Headers

Key	Required	Value	Description
Content-Type	yes	application/json	Content Type
Authorization	yes	bearer [YOUR_AUTH_TOKEN]	Access Token

Request

Field	Description	Type	M/O	Length
phoneNumber	The phone number that received the status code	String	M	16

Output

Field	Description	Type	M/O
statusCode	Status code for the response	String	M
statusMessage	status code description	string	M
accountNumber	Customer Account number	string	M
accountName	Account Name	decimal	O
accountCurrency	Account currency	decimal	O

Sample request

```
POST /api/v1/withdraw/namecheck HTTP/1.1
Host: {{baseUrl}}
Authorization: {{token}}
{
  "phonenum": "+2557031526378",
}
```



Sample successful response

```
{
  "statusMessage": "Successful",
  "statusCode": "00",
  "successful": true,
  "result": {
    "accountNumber": "32133213***343424",
    "name": "Sample account name",
    "accountCurrency": "USD"
  }
}
```

Sample failure response

```
{
  "statusMessage": "Could not find account number mapped to the phone number",
  "statusCode": "01",
  "successful": false,
  "result": {}
}
```



c. Withdrawal API

This service will be used to send the transactional withdrawal request. This call is asynchronous and there a callback will be sent

HTTP Version	HTTP/1.1
API TYPE	REST/JSON
HTTP Method	POST
End-point	{{baseUrl}}/api/v1/withdraw/transaction
Mode	Asynchronous

Headers

Key	Required	Value	Description
Content-Type	yes	application/json	Content Type
Authorization	yes	bearer [YOUR_AUTH_TOKEN]	Access Token

Request

Field	Description	Type	M/O	Length
requestID	Unique requestID	String	M	13
amount	Amount to withdraw	String	M	10
phoneNumber	The phone number to withdraw from	String	M	13
cashCode	Cashcode sent to the customer	String	M	6
providerID	The consumer provider	String	M	10
serviceID	The service that being consumed. In this case Agent withdrawal	String	M	15
transactionType	Transaction type. Use withdrawal	String	M	10
countryID	ISO format for the country	String	M	5
Currency	ISO format for the transaction currency	String	M	3
Narration	Transaction Narration	String	O	30

Output

Field	Description	Type	M/O
statusCode	Status code for the response	String	M
statusMessage	status code description	string	M

Callback Output

Field	Description	Type	M/O
statusCode	Status code for the callback	String	M
statusMessage	status code description	string	M
internalReferenceNumber	Transaction reference number from corebanking	String	M
telcoRefequestID	Similar to the request	String	M
additionalParams	Holds any additional parameters in key/value structure	Object	O



Sample request

```
POST /api/v1/withdraw/transaction HTTP/1.1
Host: {{baseURL}}
Authorization: {{token}}
Content-Type: application/json
{
  "requestID": "retrer767",
  "providerID": "TIGO_TZ",
  "serviceID": "AgentWithdrawal",
  "transactionType": "withdraw",
  "countryID": "TZ",
  "phoneNumber": "2557031526378",
  "callbackUrl": "http://samplecallbackurl:8080/vi/callback",
  "cashCode": "123456",
  "amount": 5000,
  "currency": "TZS",
  "naration": "Withdrawal at AGENT 10005 Dar"
}
```

Sample response

```
{
  "statusCode": "0",
  "statusDescription": "Transaction received for processing"
}
```

Sample failure response

```
{
  "statusCode": "01",
  "statusDescription": "Transaction failure. Cashcode expired"
}
```

Callback Result



```
{
  "statusCode": "0",
  "statusDescription": "Transaction successfully processed",
  "successful": true,
  "internalRefNumber": "Intref23243343343",
  "telcoRequestID": "ref1234566576",
  "additionalParams": [
    {
      "key": "Name",
      "value": "Test User"
    },
    {
      "key": "transactionCompletedOn",
      "value": "2022-03-24T07:08:13.4405602Z"
    }
  ]
}
```

d. *StatusCheck*

If the callback takes long to be sent, Telco will send a status check request

Sample request

```
POST /api/v1/withdraw/statusCheck HTTP/1.1
Host: {{baseUrl}}
Authorization: {{token}}
Content-Type: application/json
{
  "requestID": "ref12345665768",
  "amount": 50.00, //optional
  "transactionTime": "" //optional
}
```

Sample response

```
{
  "statusCode": "0",
  "statusDescription": "Transaction successfully processed",
  "successful": true,
  "internalRefNumber": "Intref23243343343",
  "telcoRequestID": "ref1234566576",
  "additionalParams": [
    {
      "key": "Name",
```



```
        "value": "Test User"
      },
    ]
  }
```

Sample failed response

```
{
  "statusCode": "03",
  "statusDescription": "Reason for transaction failure",
  "successful": false,
  "internalRefNumber": "",
  "telcoRequestID": "ref12345665768",
  "additionalParams": [
    {
      "key": "keyname",
      "value": "value"
    }
  ]
}
```



DEPOSIT TRANSACTION APIS

APIs to aid in cash deposits at the Agent.

a. Namecheck / Account lookup

HTTP Version	HTTP/1.1
API TYPE	REST/JSON
HTTP Method	POST
End-point	{{baseURL}}/api/v1/deposit/namecheck
Mode	Synchronous

Headers

Key	Required	Value	Description
Content-Type	yes	application/json	Content Type
Authorization	yes	bearer [YOUR_AUTH_TOKEN]	Access Token

Sample request

```
POST /api/v1/deposit/nameCheck HTTP/1.1
Host: {{baseURL}}
Authorization: {{token}}
{
  "accountNumber": "2011211305859",
  "accountType": "accountNumber",
  "countryId": "TZ",
}
```

Sample response

```
{
  "accountNumber": "32133213***343424",
  "accountName": "Sample account name",
  "accountCurrency": "USD"
}
```



b. Deposit Transaction API

HTTP Version	HTTP/1.1
API TYPE	REST/JSON
HTTP Method	POST
End-point	{{baseUrl}}/api/v1/deposit/transaction
Mode	Asynchronous

Headers

Key	Required	Value	Description
Content-Type	yes	application/json	Content Type
Authorization	yes	bearer [YOUR_AUTH_TOKEN]	Access Token

Request Parameters

Tag	Data Type	Required	Description
requestID	String	yes	Request unique reference number
amount	Number	yes	Transaction amount
Currency	String	Yes	Transacting currency
providerID	String	Yes	Provider consuming the API
served	String	Yes	Service consumed
callbackUrl	String	yes	Callback URL once the transaction has completed processing
destinationAccountNumber	String	yes	Destination account number either Till, account number, MSISDN linked to account
destinationAccountType	String	yes	Destination account type eg MSISDN, Till, AccountNumber
phoneNumber	String	yes	Phone Number of the User doing the transaction
transactionReference	String	N	Additional refnumber
senderName	String	N	Sender details
Narration	String	N	Narration of the transaction

Sample request

```
POST /api/v1/deposit/transaction HTTP/1.1
Host: {{baseUrl}}
Authorization: {{token}}
Content-Type: application/json

{
  "requestID": "ref1234566576",
  "amount": 1000,
  "currency": "TZS",
  "providerID": "TIGO_TZ",
  "serviceID": "agentDeposit",
```



```
{
  "callbackUrl": "http://samplecallbackurl:8080/vi/callback",
  "destinationAccountNumber": "1234567890123",
  "destinationAccountType": "accountNumber",
  "phoneNumber": "255124***778",
  "transactionReference": "32Ref2",
  "narration": "Rent payment 1234",
  "senderName": "Test User 1"
}
```

Response

Tag	Data Type	Required	Description
statusCode	String	Y	Status code 00 - success, 01/02-failed
statusDescription	String	Y	Description to the statuscode

Sample response

```
{
  "statusCode": "0",
  "statusDescription": "Transaction received for processing"
}
```

Callback Result

Tag	Data Type	Required	Description
statusCode	String	Y	Status code 00 - success, 01- pending 02-failed
statusDescription	String	Y	Description to the statuscode
successful	Boolean	Y	True or False
internalRefNumber	String	Y	Internal reference number for the transaction
telcoRequestID	String	Y	Reference to the request reference number
additionalParams	ObjectArray	Y	Holds additional information in key-value pair format

```
{
  "statusCode": "0",
  "statusDescription": "Transaction successfully processed",
  "successful": true,
  "internalRefNumber": "Intref23243343343",
  "telcoRequestID": "ref1234566576",
  "additionalParams": [
    {

```



```
        "key": "Name",
        "value": "Test User"
    }
  ]
}
```

c. StatusCheck

If the callback takes long to be sent, Telco will send a status check request

Sample request

```
POST /api/v1/deposit/statusCheck HTTP/1.1
Host: {{baseUrl}}
Authorization: {{token}}
Content-Type: application/json
{
  "requestID": "ref12345665768",
  "amount": 50.00, //optional
  "transactionTime": "" //optional
}
```

Sample response

```
{
  "statusCode": "0",
  "statusDescription": "Transaction successfully processed",
  "successful": true,
  "internalRefNumber": "Intref23243343343",
  "telcoRequestID": "ref1234566576",
  "additionalParams": [
    {
      "key": "Name",
      "value": "Test User"
    },
  ],
}
```



Sample failed response

```
{
  "statusCode": "03",
  "statusDescription": "Reason for transaction failure",
  "successful": false,
  "internalRefNumber": "",
  "telcoRequestID": "ref12345665768",
  "additionalParams": []
}
```

STATUS CODES FOR TRANSACTION STATUSES

statusCode	description
00	Transaction posted successfully
01	Transaction is pending
02	Transaction failed with a specific error
03	Transaction failed with an unspecified error
04	Transaction not successful in CBS
05	Transaction failed with an unspecified error
06	Cross currency transaction not allowed
-1	Transaction not found

HTTP RESPONSE CODES

Http Response Code	Response Code	Description
202	0	Request processed successfully
400	1	Bad request
401	1	Unauthorized request
403	1	Forbidden
404	1	Not found
500	1	Internal server error



OMNI – GENERATE CASHCODE / VALIDATE CASHCODE

Link: [Generate Cashcode Service](#)

Link: [Validate cashcode](#)



5. SECURITY ARCHITECTURE

- **Message signing** – Equity Bank will use her private key to sign the API messages, and his public key is used to read the signature. This will help to ensure data integrity, message authentication, and non-repudiation.
- **Oauth 2.0** – Auth 2.0 is an authorization protocol that gives an API client limited access to user data on a web server based on access token granted to the API client
- **IP whitelisting** – Partners and third party system configuration (IP Address) is captured and validated
- **Connection over VPN** – Connection between Equity bank and MTN will be through a VPN
- **Role Based Access Control** – RBAC is implemented to segment users and their privileges
- **User Activities** – User (API) calls will be logged which will enhance audit trail
- **Error Messages** – Error message type and their description will be defined



6. RESOURCES REQUIRED

The actualization of this design requires

- Quality assurance to assist with testing of the design
- Developers from both Equity bank and TigoPesa to build the required functionality
- Architects to provide guidance where necessary.

7. SERVICES SLA

Telco to Bank (Deposit, Withdrawal and Namecheck call)

- TigoPesa – Equity Bank: Proposed architecture will be synchronous and therefore response from the bank is expected in not more than 2 seconds.
- However, in instances the call is to implemented in asynchronous, a response will be issued within 2 seconds of request and a result expected in not more than 15 seconds
 - If no response within 15 seconds, a transaction status call is made to provide the status of the transaction
 - Pending transactions will be defined maximum number of retries to happen before the transaction is marked as failed

Bank to Telco (B2Wallet & Namecheck)

Proposed architecture is synchronous calls and therefore the average expected response time is 2 seconds. On instances the calls are changed to asynchronous, the flow will follow the above SLA.

8. REFERENCED DOCUMENTS

Document	Attachment	Comments
3 rd party utility API specification	Tigo Pesa API Spec	To update once updated API is shared
Business case	Business Case	
BRD.doc	BRD Document	Business requirements document